# Literature Review: Outline

## Define

**Property value, market price, and BIR zonal values.** Market value is the best estimate of what a property should sell for in normal conditions. Market price is the amount buyers and sellers actually agree on. In the Philippines, BIR zonal values are government assessments for tax. They often lag the market, so we will use zonal value both as a predictor and as a benchmark to compute the valuation gap, which is market price minus zonal value (Domingo & Fulleros, 2005).

**Hedonic pricing (linear and multiple regression).** This models price from property and location features, so we can see how much each feature contributes to price (Rosen, 1974; Malpezzi, 2003). It is a clear baseline and can be extended with simple spatial terms.

**Decision trees and ensembles.** A decision tree predicts price by splitting the data on useful features. Random forests (Breiman, 2001) and gradient boosting (Friedman, 2001) combine many trees to improve accuracy, with some loss in transparency.

**Evaluation metrics.** We will report mean absolute error, root mean squared error, mean absolute percentage error, and R-squared, using holdout or time-aware cross-validation (Hyndman & Athanasopoulos, 2021).

## Characterize

**Philippine determinants and data.** Local attributes matter. In Pangasinan, models that used BIR zonal values, the BSP Residential Real Estate Price Index, and a construction cost index achieved high accuracy with random forest and beat multiple linear regression (Viray, 2023). In Metro Manila, hedonic models with quality and environmental indicators, plus spatial spillovers, explained rent differences (Dann et al., 2020).

**International modeling patterns.** Across many housing datasets, boosted trees such as gradient boosting and XGBoost often outperform linear models (Sharma et al., 2024; Weng, 2022). This supports testing both linear and tree-based models on the same Philippine data.

## Relate

**Zonal values and market prices.** Zonal values are a useful baseline but need to be combined with transactions and richer features. Philippine evidence shows that adding indices and costs can improve predictions (Domingo & Fulleros, 2005; Viray, 2023).

**Method comparisons.** We will run head to head tests of multiple linear regression against decision trees and simple ensembles, and measure the incremental value of adding zonal values by looking at changes in MAE and R-squared.

# Anticipate

**Significance.** A simple and reproducible pipeline that combines zonal values, market prices, GIS proximity features, risk layers, and macro indicators can produce nowcasts and 3 to 12 month forecasts, map valuation gaps, and help appraisers, brokers, and local governments.

**Research gap.** Philippine studies rarely combine property level prices with zonal values, GIS proximities, risk, and macro context while doing short horizon forecasts. Our framework fills that gap.

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# Literature Map